

Q1. What is the difference between valve-regulated lead-acid (VRLA) and sealed maintenance-free (SMF) batteries?

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A1. Both batteries are sealed and maintenance-free. VRLA batteries are low-maintenance lead-acid rechargeable batteries. These are often called sealed lead-acid batteries or recombinant batteries. They have a pressure-relief valve that activates when the battery is recharged at a high voltage (typically greater than 2.30V per cell).

VRLA cells may be made of flat plates similar to a conventional flooded lead-acid battery or made in a spiral roll form to make cylindrical cells. The battery utilises a separator of glass fibre, which isolates the negative and positive plates and acts as a blotter to absorb all the electrolyte within the cell.

VRLA batteries are available in two categories—absorbed glass mat (AGM) and gel (gel cell). AGM batteries are just like flooded lead-acid batteries, except that the electrolyte is held in glass mats, as opposed to freely flooding the plates. Very thin glass fibres are woven into a mat to increase the surface area enough to hold sufficient electrolyte on the cells for their lifetime.

Gel battery is a VRLA battery with a gelled electrolyte; sulphuric acid is mixed with silica fume to make the resulting mass gel-like and immobile. The separator is not relied upon to absorb the electrolyte, since the electrolyte is gelled. It strictly performs the function of separating and resisting the development of shorts between the plates.

As the electrolyte is blotted or gelled, gel batteries are spill-proof and can be mounted upside down. This makes them usable and rechargeable in any position.

The oxygen produced at the positive plates largely recombines with the hydrogen on the negative plates, creating water and thus preventing water loss. These batteries therefore do not require regular addition of water to the cells making them maintenance-free.

The valve is a safety feature in case the rate of hydrogen production

becomes dangerously high. In flooded cells, water must be periodically added as the gases escape before having a chance to recombine.

The use of special sealing epoxies, tongue and groove case, cover construction and long-sealing paths for posts and connectors ensures that the VRLA battery offers exceptional leak resistance and can be used in any position. These batteries do not pollute air, unlike lead-acid batteries that give out lead fumes polluting the surrounding air.

SMF batteries are fully sealed, safe and maintenance-free. These batteries do not have a pressure-relief valve and do not need any water topping.

Q2. When I connected one end of a 1-mega-ohm resistor to the negative (or positive) terminal of a 9V battery, the voltage between the other end of the resistor and the positive (or negative) terminal of the battery was measured as 4.85 volts. When I varied the resistance, the measured voltage decreased with the increase in resistance. What is the reason for this decrease in voltage? Is there any relationship to find the measured voltage using supply voltage and resistor value? In many op-amp circuits, the input voltage is given to the inverting and non-inverting terminals through a resistor. What is the purpose of connecting this resistor?

Packia Selvan
Through e-mail

A2. The answer to the first part is based on the arrangement of connection you have described (refer Fig. 1). The voltmeter has been replaced with its resistance R_M for analysis of the circuit. It is a series circuit comprising R_S (internal resistance of the battery), R (1 M Ω) and R_M . The value of R_M is determined by calculating current in the circuit.

Taking the terminal voltage of the battery equal to 9V (neglecting drop across R_S), the drop across R (1 M Ω) is 4.15V (9-4.85V) and current is 4.15 μ A. The value of R_M therefore will be 4.85/4.15=1.16 M Ω .

The input impedance of a voltmeter is very high and current drawn

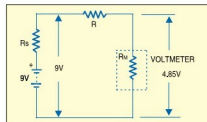


Fig. 1: Voltage measurement

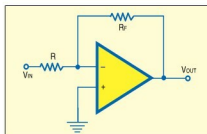


Fig. 2: Non-inverting op-amp

very small, which does not affect the voltage being measured. When you increase resistance R , say, to 2 M Ω , the total resistance becomes 3.16 M Ω . The current in the circuit reduces to $9/3.16=2.8 \mu$ A. Being a series circuit, this current also flows through R_M . The reading of the voltmeter now will be $2.8 \times 1.16=3.24$ V. The increase in resistance results in a drop in the voltage, which you too have observed.

When the op-amp amplifier works in comparator mode, resistors are connected to form a voltage divider network to provide the reference voltage and the voltage which is being compared with the reference voltage. However, when the device operates in inverting/non-inverting amplifier mode, the input voltage is fed through the resistor so that the variation can be achieved at the terminal of the device. In case of a non-inverting amplifier, the gain factor is $-R_f/R$ as shown in Fig. 2. R_f is a feedback resistor. If R is zero, gain will be infinite and output voltage will not be related to the input voltage by the gain factor, which is expected from an amplifier circuit.

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